Generative Artificial Intelligence

Introduction





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http://www.basantajoshi.com.np

https://scholar.google.com/citations?user=iocLiGcAAAAJ

https://www.researchgate.net/profile/Basanta_Joshi2

Manish Awale, PhD Candidate Department of Electronics and Computer Engineering, Pulchowk Campus, Institute of Engineering, Tribhuvan University

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Senior Computer Vision Engineer, EKbana Solutions Pte. Ltd

About me

Current Affiliation

Development (LICT)
http://lict.ioe.edu.np/

Education

- Bachelor of Electronics and Communication Engineering, IOE 2005
- MSc in Information and Communication Engineering, IOE, 2008
- Doctor of Engineering, Osaka Sangyo University, Japan

Industry Experience

- Assistant Professor, Department of Electronics and Computer Engineering, Pulchowk Campus
- https://pcampus.edu.np/
- Deputy Director, Center for Applied Research & Development(CARD) https://card.ioe.edu.np/ • Member, Laboratory for ICT Research and
- Senior Software Engineer, D2hawkeye, Nepal • Research Consultant, LogPoint, Nepal
- System Administration and Web development, Japan • IT Consultant In Various National & International Projects

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Interests and Research Area

- Motion Tracking & Video Analytics
- Nepali Language Processing
- Medical Data Analytics
- Network & Edge Analytics

Use of Big data Analytics in above mentioned Areas

2013

Subjects Taught

- Artificial Intelligence, Machine Learning, Deep Learning, Big Data Analytics, Image Processing, Audio & Speech Processing Business Analytics e.t.c

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Course Description:

- This course offers an in-depth introduction to Generative Artificial Intelligence (GenAI), combining foundational programming skills with cutting-edge developments in large language models and AI

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application frameworks.

- Students will begin by mastering Python and essential data libraries, then delve into the architecture of LLMs, advanced prompting techniques, and tools like LangChain for building complex generative systems.
- The course also covers Retrieval Augmented Generation (RAG), ethical considerations in AI, and emerging topics such as multimodal and agentic systems.
- By the end of the course, students will be equipped to design, build, and critically evaluate GenAI applications across a range of contexts.

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Course Objectives:

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1. Build foundational skills in Python and essential data libraries for AI development.
2. Understand the architecture, training, and evolution of large language models in generative AI.
3. Apply prompt engineering, LangChain, and RAG techniques to build and optimize GenAI applications.
4. Analyze ethical, social, and practical implications of GenAI in real-world contexts.

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Course Contents

Develop foundational Python skills and familiarize with core data libraries and tools useful for generative AI development.	**Unit 1: Introduction to Generative AI and Python Fundamentals** **(7 hrs)** 1.1 Evolution of AI and Emergence of Generative AI 1.2 Python Syntax and Data Types 1.3 Control Structures and Functions in Python 1.4 Introduction to NumPy for Numerical Operations 1.5 Pandas for Data Manipulation and Analysis 1.6 Data Visualization with Matplotlib 1.7 Introduction to FastAPI for API development

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Familiarize with the architecture, types, and training methodologies of modern large language models.	**Unit 2: Foundations of Large Language Models (6 hrs)** 2.1 Transformer Architecture and Attention Mechanisms 2.2 Key LLM Types: GPT, BERT, LLaMA, 2.3 Training Methodologies: Pre-training and Fine-tuning

Course Contents

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Specific Objectives	Contents
Implement the zero-shot and few shot prompts to build effective Gen AI applications.	**Unit 3: Prompt Engineering Fundamentals (8 hrs)** 3.1 Introduction to Prompt Engineering 3.2 Anatomy of Effective Prompts 3.3 Zero-shot Learning Techniques 3.4 Few-shot Learning and Examples 3.5 Chain-of-Thought Prompting 3.6 Role Prompting and System Instructions 3.7 Prompt Templates and Frameworks 3.8 Case Study: Interactive Prompt Workshop with LLM models
Build complex LLM-based applications using the LangChain framework by leveraging chains, memory, and tool integration.	**Unit 4: LangChain Framework and Applications (8 hrs)** 4.1 LangChain Architecture and Components 4.2 Chains and Sequence Processing 4.3 Prompt Templates and LLM Integration 4.4 Memory Systems in LangChain 4.5 Agents and Tools Implementation 4.6 Document Loading and Text Splitting 4.7 Building Conversational Applications 4.8 Case Studies of LangChain Applications

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Implement context-aware LLM systems using vector embeddings, similarity search, and evaluation metrics in RAG pipelines.	**Unit 5: Retrieval Augmented Generation (RAG) (8 hrs)** 5.1 Introduction to Retrieval Augmented Generation 5.2 Understanding Tokens and Token Limitations 5.3 Vector Embeddings: Principles and Generation 5.4 Vector Databases: Pinecone, Chroma, 5.5 Chunking Strategies and Metadata 5.6 Similarity Search and Retrieval Mechanisms 5.7 Hybrid Search Methods 5.8 Re-ranking Techniques 5.9 Evaluation Metrics: BLEU, Context precision, recall, faithfulness, relevance
Analyze ethical challenges in generative AI and apply responsible AI practices across the model lifecycle.	**Unit 6: Ethical and Responsible AI (4 hrs)** 6.1 Introduction to Ethics in Generative AI 6.2 Sources and Impacts of Bias in LLMs 6.3 Fairness, Representation, and Social Harm 6.4 Explainability and Transparency Challenges 6.5 Model Misuse, Hallucinations, and Jailbreak Risks 6.6 Data Privacy, Consent, and Legal Considerations (e.g., GDPR, AI Act)
Familiarize students with emerging trends in generative AI, including multimodal systems, agentic models,	**Unit 7: Advanced Topics in Generative AI (4 hrs)** 7.1 Fundamentals of Multimodal AI 7.2 Agentic AI Systems 7.3 Emerging Research Directions 7.4 Self-hosting LLMs 7.5 Finetuning LLMs

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and advanced fine-tuning techniques.	

References

- Prompt Engineering Guide by West, H., & Bommasani, R. (2023). GitHub. https://github.com/dair ai/Prompt-Engineering-Guide
- Language models are few-shot learners by Brown, T. B., et al. (2020). arXiv preprint arXiv:2005.14165. https://arxiv.org/abs/2005.14165
- Retrieval-augmented generation for knowledge-intensive NLP tasks by Lewis, P., et al. (2020). arXiv preprint arXiv:2005.11401. https://arxiv.org/abs/2005.11401
- LLaMA: Open and efficient foundation language models by Touvron, H., et al. (2023). arXiv preprint arXiv:2302.13971. https://arxiv.org/abs/2302.13971
- Python for Data Analysis: Data Wrangling with pandas, NumPy, and Jupyter by McKinney, W. (2022). https://wesmckinney.com/book/

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- FastAPI Cookbook: Develop high-performance APIs and web applications with Python (2023) • The Big Book of Generative AI by Databricks (2025).
- Rothman, D. (2024). RAG-Driven Generative AI: Build custom retrieval augmented generation pipelines with LlamaIndex, Deep Lake, and Pinecone. Packt.

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Teaching Methodology

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• Lecture includes slides, video and discussions • Students should submit assignments including programming assignments/Quizzes in google classroom

• Students should do presentation upload in google classroom

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- Google classroom
  - **Class code →** pgwi63jy

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Intro to the course

Following questions we aim to answer together:

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- How does generative AI work, and what technologies make it possible? • How do large language models understand, learn, and generate human-like content?
- How can prompt engineering and AI frameworks be used effectively to build intelligent applications?
- How can retrieval systems and external knowledge improve the accuracy and usefulness of AI?
- How will generative AI affect society, ethics, and the future of technology?

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The debut of mass market LLMs

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- ChatGPT released November 30, 2022
  - Not the first publicly available large language model (LLM), but heralded the beginning of widely used LLMs in non technical hands

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The debut of mass market LLMs

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- ChatGPT released November 30, 2022
  - Not the first publicly available large language model (LLM), but heralded the beginning of widely used LLMs in non technical hands

- Within 5 days it had 1 million users.
- On December 3, Aaron Levie, CEO of Box tweeted:



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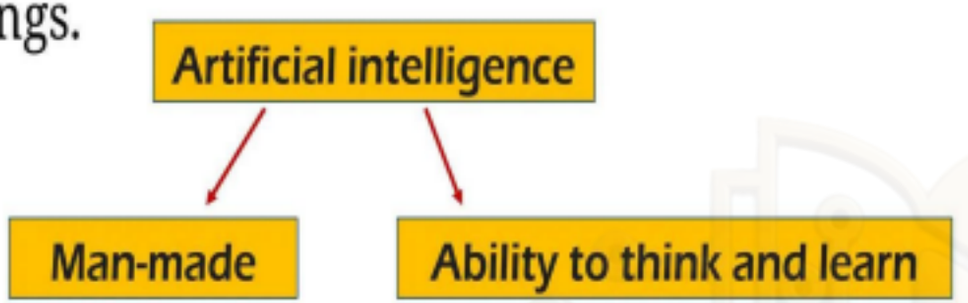
Definition of AI

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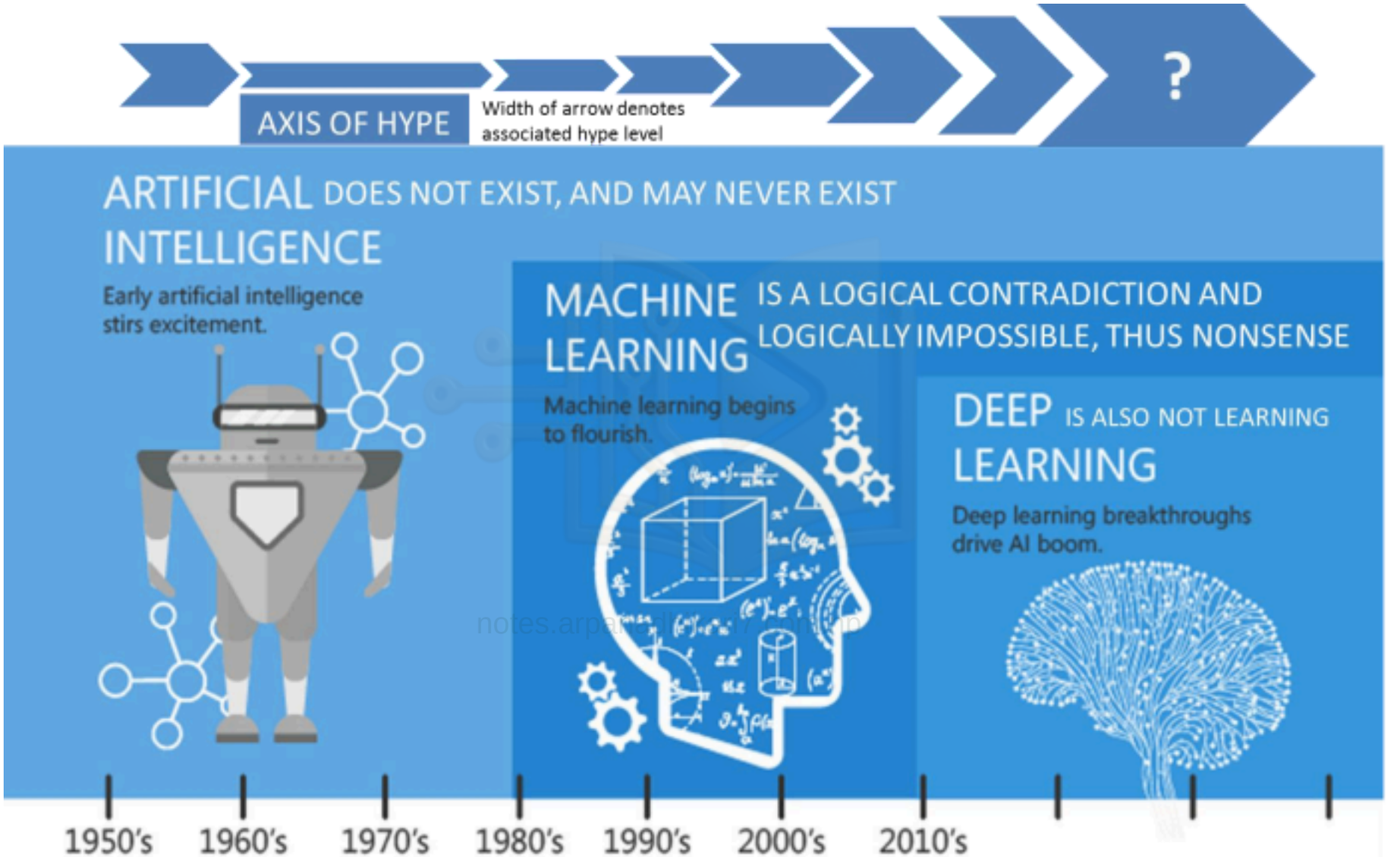
- **Artificial Intelligence** is the ability of a machine to think, understand, learn, and work like human beings.



- AI focuses on three cognitive skills – reasoning, learning and self-correction.

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Since AI has continued to not exist since it was first posited as possible, hype has become the only means of maintaining interest and funding in AI. By creating smaller subsets of something that has

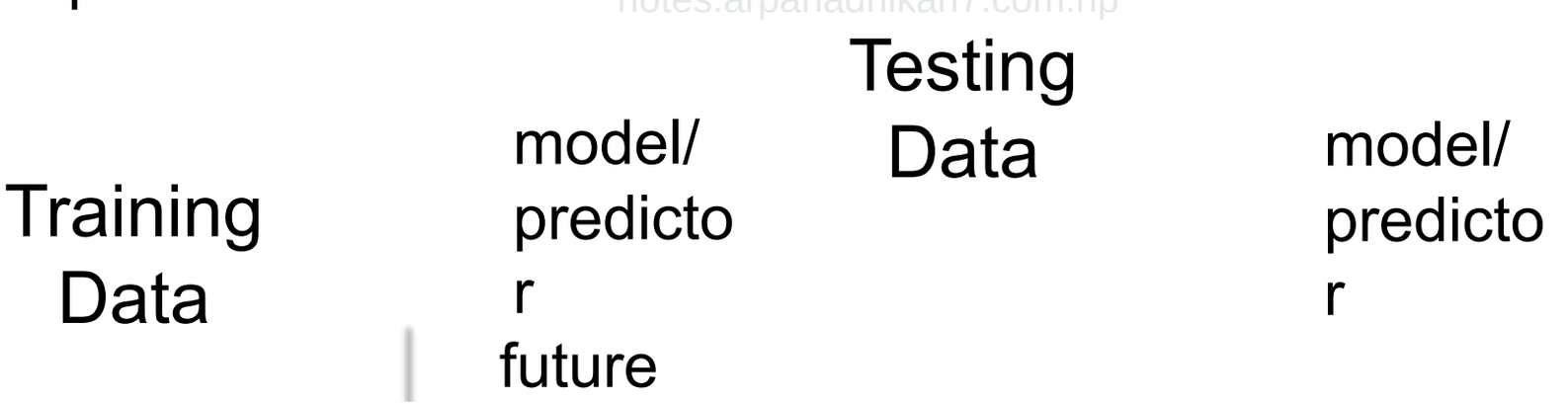
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Machine Learning is…

Machine learning is about predicting the future based on the past.

-- Hal Daume III

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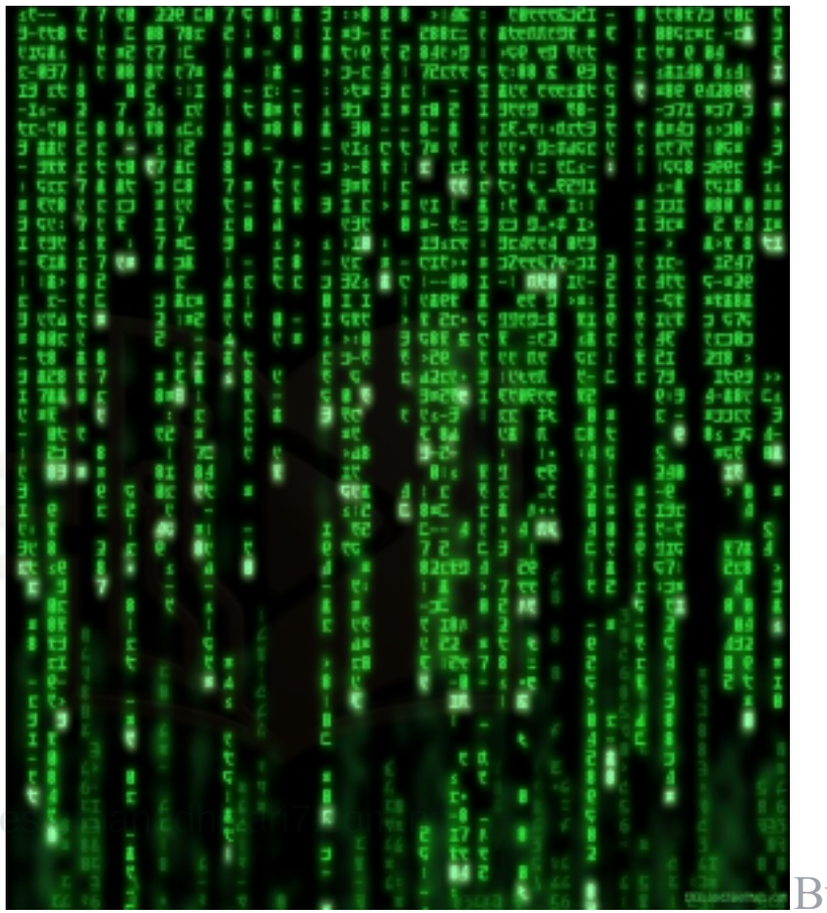


Why Machine Learning is Hard?

You See Your ML Algorithm Sees A Bunch of Bits

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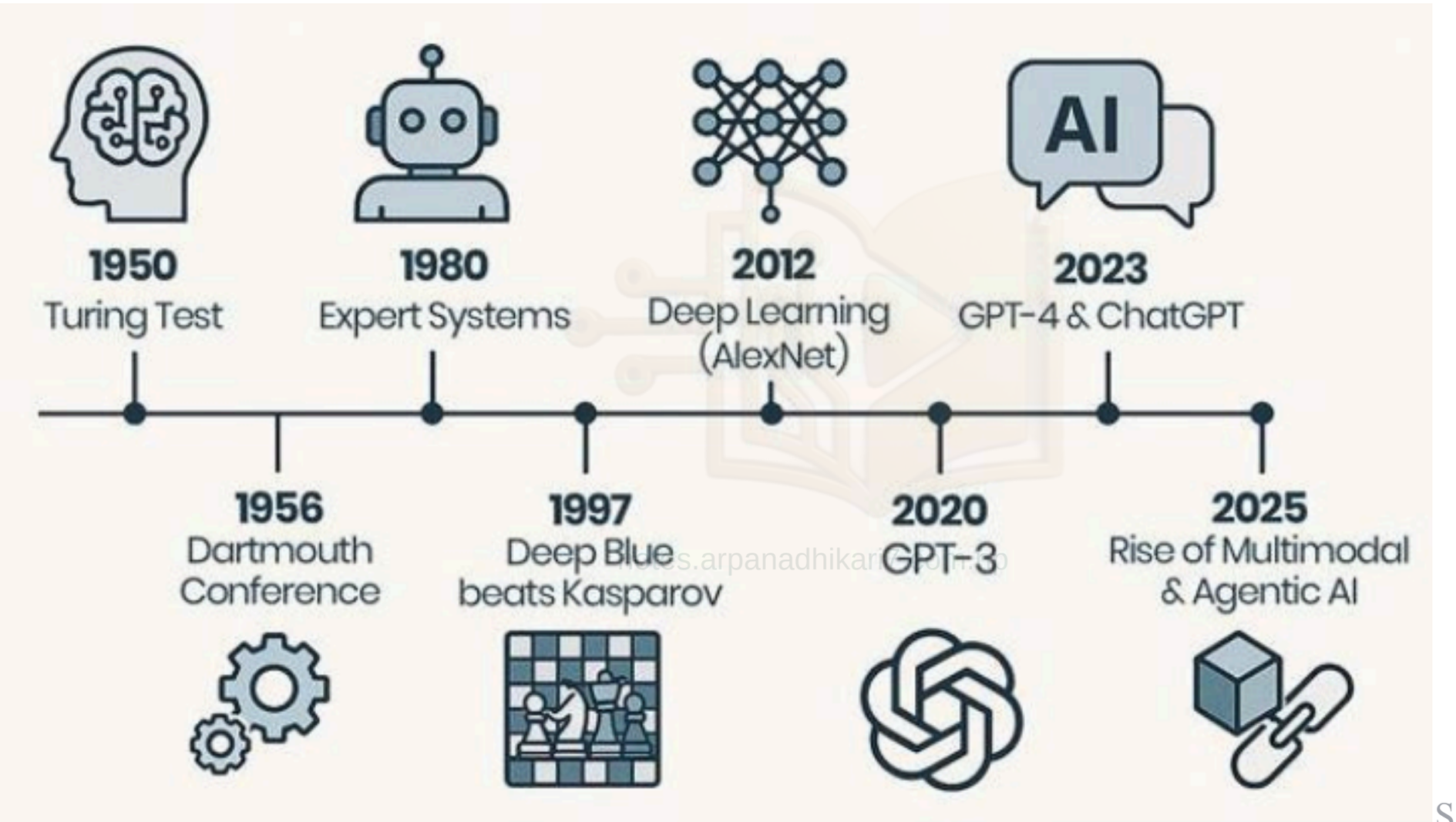
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History of Artificial Intelligence

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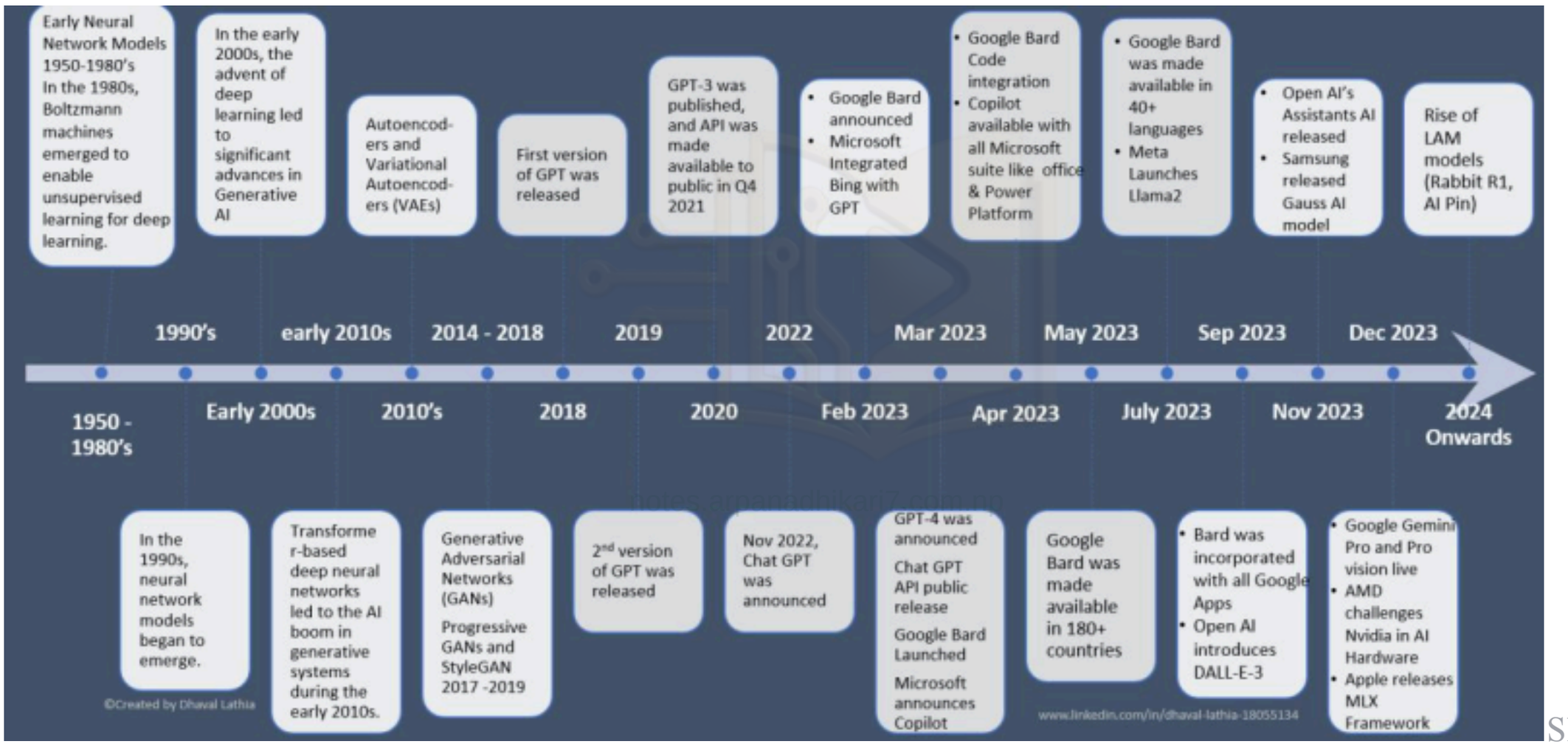
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Basanta Joshi, PhD

Major Milestones of Generative AI

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AI Vs GAI

- **Artificial Intelligence**: a broad term that refers to any technology that is capable of intelligent behavior. This can include a wide range of technologies, from simple algorithms that can sort data, to more advanced systems that can mimic human-like thought processes.

- **Generative Artificial Intelligence**: a specific type of AI that is focused on generating new content, such as text, images, or music. These systems are trained on large datasets and use machine learning algorithms to generate new content that is similar to the training data. This can be useful in a variety of applications, such as creating art, music, or generating text for chatbots.

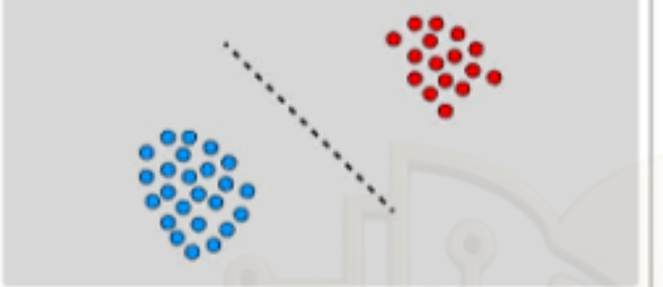
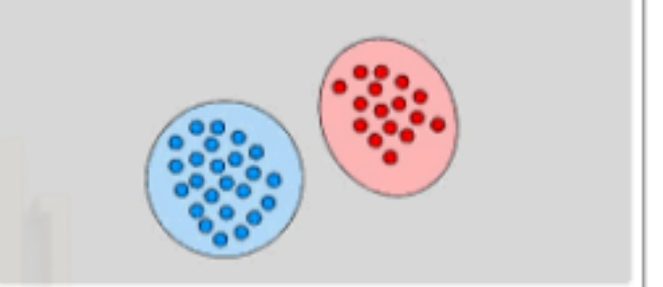
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Source: https://www.antler.co/blog/generative-ai

Discriminative models Vs.Generative

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models

Discriminative Models	Generative Models
	
Learns the decision boundary between classes	Learns the input distribution
Maximizes the conditional probability: P(Y|X)	Maximizes the joint probability: P(X, Y)
Directly estimates P(Y|X)	Estimates P(X|Y) to find P(Y|X) using Bayes' rule
Cannot generate new data	Can generate new data
Specifically meant for classification tasks	Typically, they are NOT used to solve classification tasks
Discriminative models don't possess generative properties	Generative models possess discriminative properties

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Generative AI

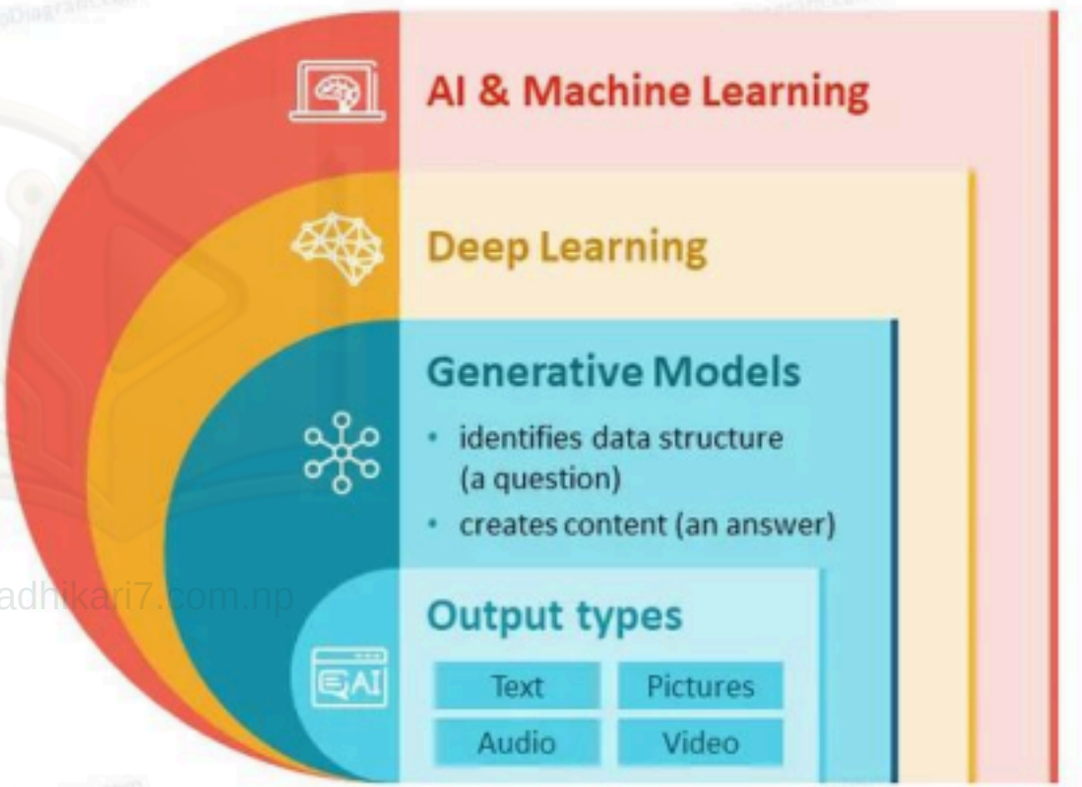
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Tools like ChatGPT fall under the category of generative AI, which is a subset of the broader field of artificial intelligence.

ChatGPT is a generative AI-powered natural language processing tool known for its high consistency and relevance.

- Write your text here. Put your text here. Add a detailed description. Write your text here. Put your text here. Add a detailed description.
- Write your text here. Put your text here. Add a detailed description. Write your text here. Put your text here. Add a detailed description.

Generative AI systems can produce new data based on learned patterns to create images, audio narrations, or text.



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GAI Tools and Outputs

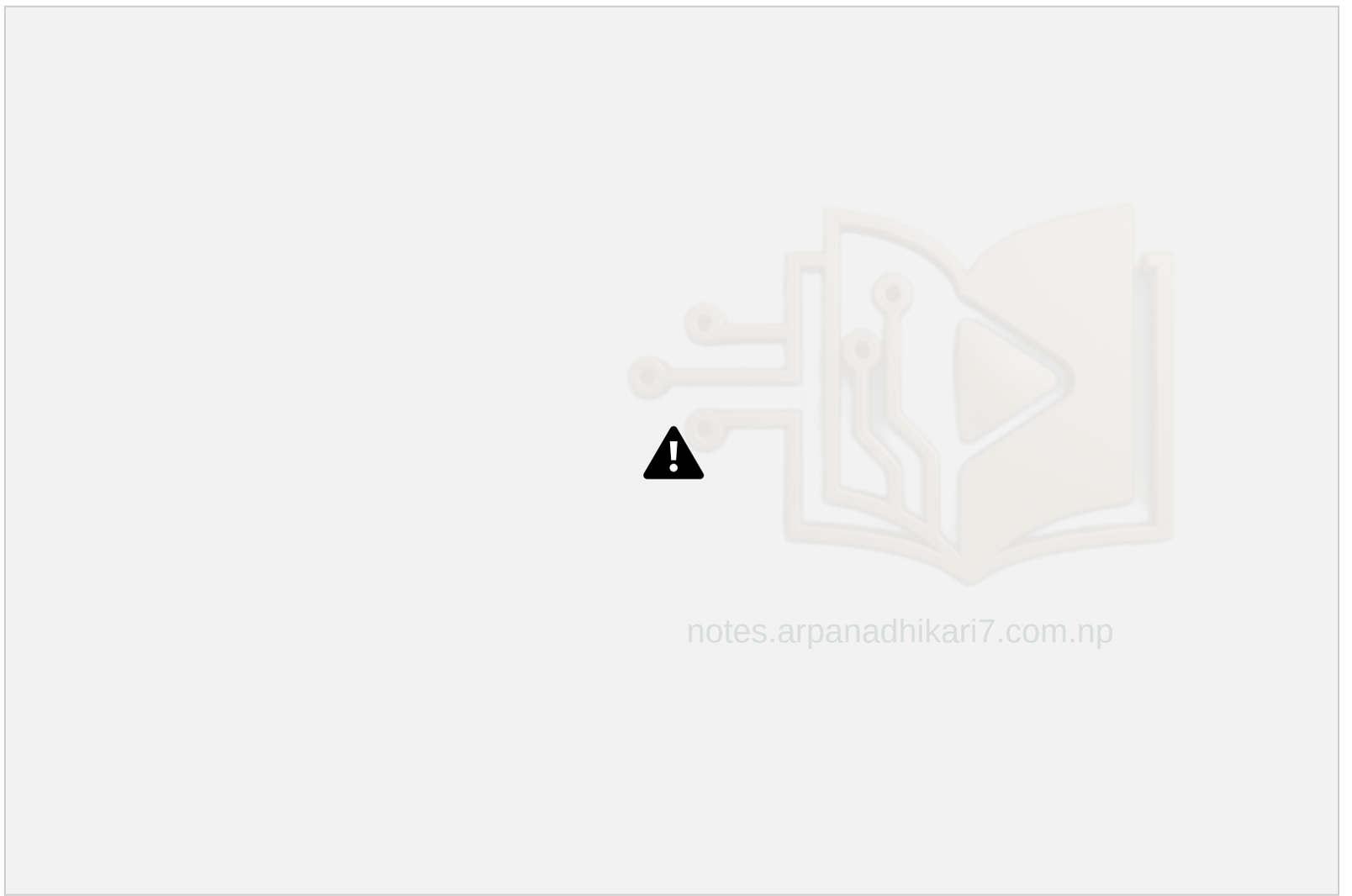
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Popular AI tools

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Best AI platforms for supporting teaching and learning

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How does GAI differ from "traditional" AI?

- Prediction can be very useful, particularly when we need to **allocate limited resources**.

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How does GAI differ from "traditional" AI?

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- Prediction can be very useful, particularly when we need to **allocate limited resources**.

  - **Building inspectors** can prioritize buildings that have elevated fire risk.
  - **Tax auditors** can prioritize individuals who have high likelihood of underpayment.
  - **Hospitals** can provide transportation for patients at high risk of not appearing at their appointments.

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What's new about Generative AI?

•The latest generation of **generative artificial intelligence** can produce novel content — including **essays**, **computer code**, and **artwork** — that is often hard to distinguish from materials produced by humans.

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Your uses of Generative AI (GAI)

1. Creative & Artistic Uses (e.g., generation of art, poetry, music)

2. **Academic & Research Assistance** (e.g., project brainstorming, citation generation, summarizing articles)

3. **Application in Coding & Data Visualization** (e.g., crafting Python code, creating graphs and unique visualizations)

4. **Drafting Documents & Content Generation** (e.g., drafting invitations, advocacy plans, internship applications)

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Gen AI Applications

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Generative AI

- According to a Goldman Sachs report, 3 key features of the technology relative to traditional AI that make it potentially impactful and quickly adopted:

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Generative AI

- According to a Goldman Sachs report, 3 key features of

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the technology relative to traditional AI that make it potentially impactful and quickly adopted:

1. **Generalized** rather than specialized use cases
2. **Generative** rather than descriptive - can produce original results indistinguishable from human output
3. **Approachable** interfaces that both understand and respond with natural language, images, audio and video

- So how can we use it?

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Use of GAI for Admission Process

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✓ How can **applicants** use generative AI? ✓ How can the **Admissions Office** use generative AI?

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How can applicants use generative AI?

1. **Writing Assistance:** Essay draft, proofreading, and editing

2. **Interview and Test Prep:** Preparing for interviews (and standardized tests like GRE/GMAT/TOEFL
3. **Resume and CV Optimization:** Rewriting, including key experiences, listing achievements, and embedding values
4. **Research and Information Gathering:** Understanding program requirements, comparing programs, researching faculty and funding opportunities

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Use of GAI for Admission Process

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✓ How can **applicants** use generative AI? ✓ How can the **Admissions Office** use generative AI?

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Typical factors considered in admissions (in U.S.)

- Previous grades

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- Admission test scores (SAT, ACT, GRE, etc.)
- Recommendation letters
- Essays
- Diversity

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Typical factors considered in admissions (in U.S.)

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- Recommendation letters
- **Essays**
- Diversity

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How can the admissions office use generative AI?

1. **Screening and Filtering Applications** (Based on minimum requirements, GPA, resume checks, identifying AI-written essays, and initial screening)
2. **Enhancing Diversity and Fairness** (Assessing diversity and minimizing bias, ensuring a diverse cohort, and incorporating diversity in application processing)
3. **Summarization and Analysis of Essays and CVs** (Summarizing essays and resumes, detecting plagiarism, evaluating essay quality, and highlighting strengths)
4. **Improving Communication and Information Accessibility** (Creating chatbots for FAQs, drafting personalized emails, and producing promotional material)

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Large language models

“Large language models” [LLMs] are algorithms [like ChatGPT] that mimic human language.

At a high level, LLMs are built in three steps:

1. Collect a large **corpus of text** generated by humans;

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2. Based on this text, **learn to predict** the next word in any given sequence of words;

3. **Fine-tune** the language model to **align** with desired behavior.

Step 1: Collect *a lot* of text

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- LLMs are typically trained on a massive dataset of web pages, online forums, books, and other content — comprising **trillions of words of text**.

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Step 2: Learn to predict words

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- By analyzing this massive corpus of text, an LLM learns to **predict the next word** in a sequence of words. By repeating this process over and over again, it can generate long strings of coherent text.

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Step 2: Learn to predict words

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- By analyzing this massive corpus of text, an LLM learns to **predict the next word** in a sequence of words. By repeating this process over and over again, it can generate long strings of coherent text.

- To see this process in action, we'll collectively complete the following sentence one word at a time!

[Humans mimicking machines mimicking humans!]

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Human language	automobile …	**surprisingly** very	**predictable.** …
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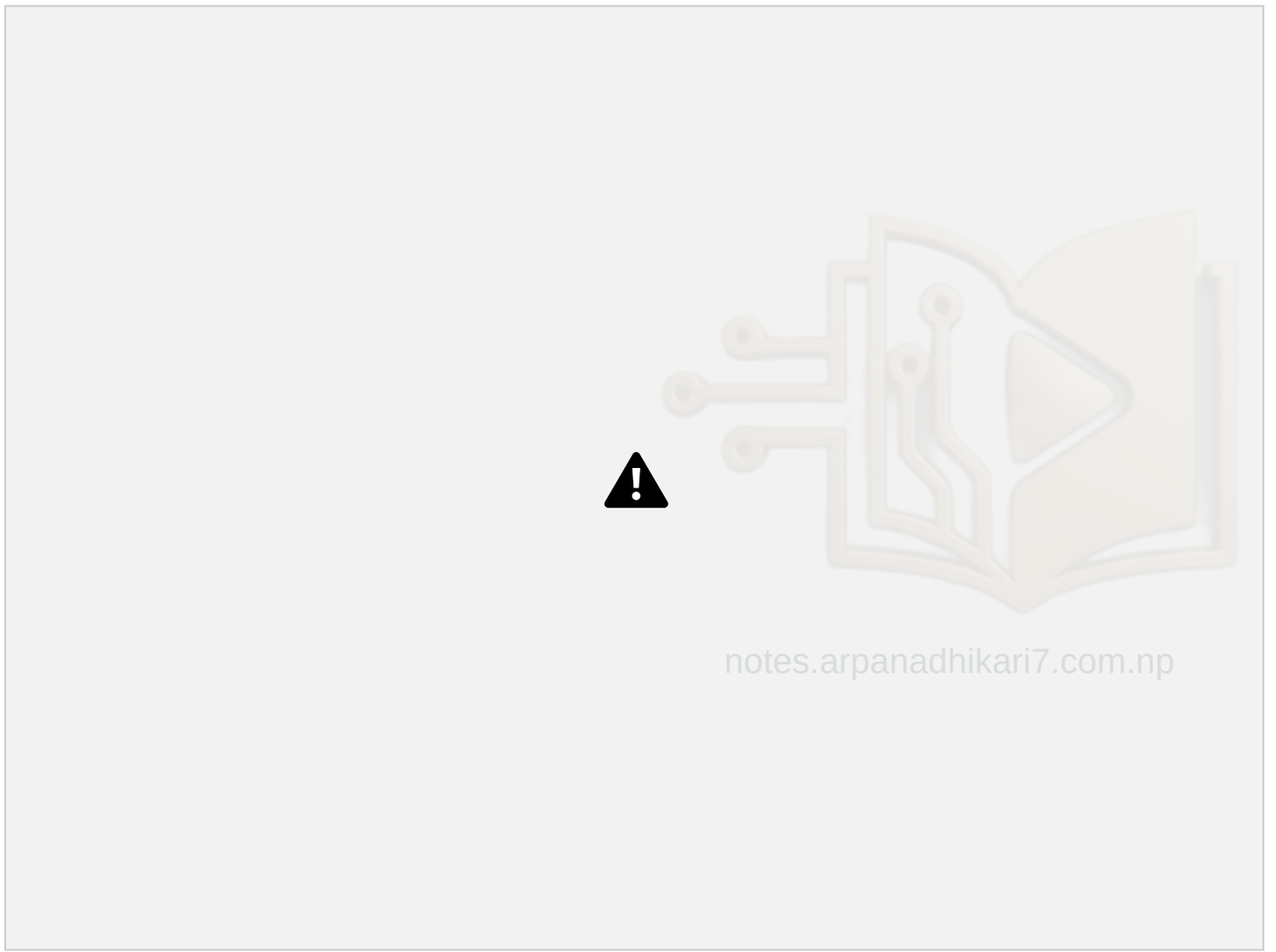
	is	banana	**versatile.** big
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Autocomplete on steroids

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Autocomplete on steroids

•Adult native-English speakers have a typical vocabulary size of about 30,000 words. The number of ways to string 10 of those words together is:

**•1,000,000,000

Autocomplete on steroids

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•Even relatively brief utterances are typically unique — they've never been said before and they'll never be said again. So we have no hope of finding exactly matching phrases in the data.

•We instead need to build a statistical **model** of language.
[i.e., a large language model — more on this next time.]

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Prediction is knowledge

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- Large language models can **encode knowledge** through **word prediction**, since correct answers typically involve **more probable** sequences of words than incorrect answers.

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Prediction is knowledge

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- Imagine trying to predict the next word in the following sequence of words:
“The capital of Australia is _____”



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Prediction is knowledge

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- Imagine trying to predict the next word in the following sequence of words: **“The capital of Australia is _____”**

- The next word is **more likely** to be “**Canberra**” than “**Sydney**” or “**Melbourne**”. [And much more likely than “**banana**”!]

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Prediction is knowledge

- To what extent can human knowledge, reasoning, and creativity be represented as a prediction problem that a language model can in theory

emulate?

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Evolution of large language models

- Before 2017, there were a few large language models that were large compared to capacity available now.
- 1990s: IBM created statistical models for machine translation. Language models relied on math-based probability (e.g., predicting words based on frequency)
- 2001: N-gram models (predicting words based on previous words) trained on 300 million words were the best available.
- As the internet grew, researchers used web text to train models, leading to better results.The more text available, the better the model performed—so internet data was used to improve them.

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Evolution of large language models

- 2009: Statistical models replaced rule-based models because they handled more data efficiently.

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- After neural networks became dominant in image processing



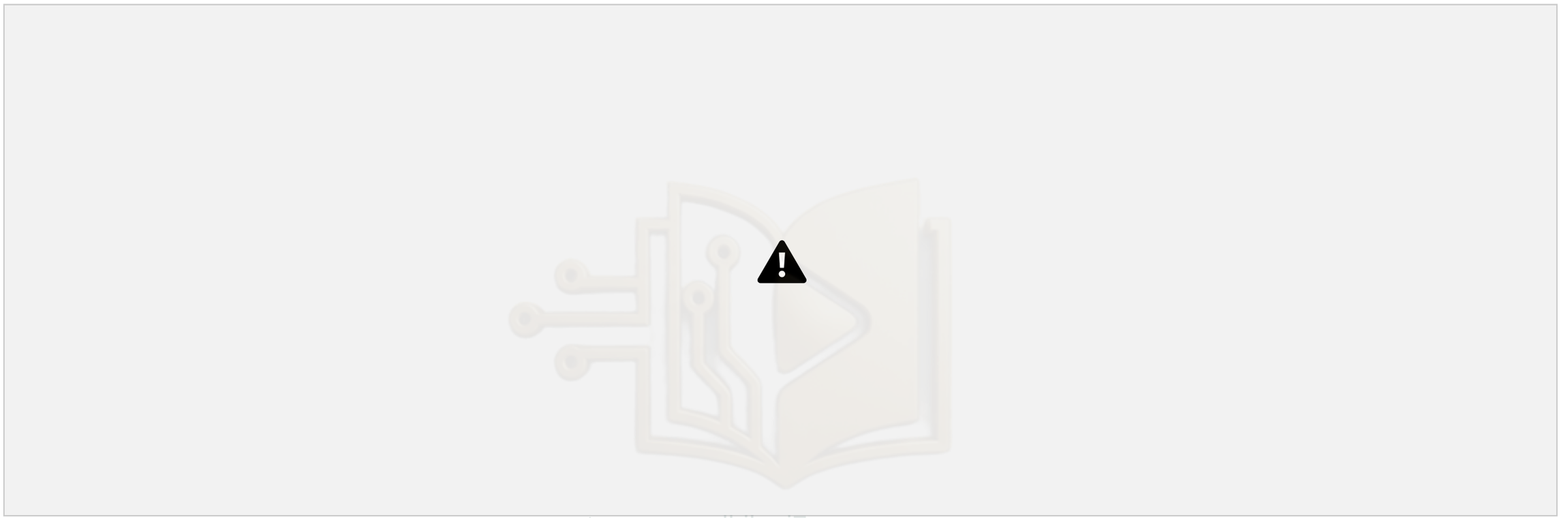
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around 2012, they were applied to language modelling as well. Traditional language models relied on probability but deep learning models started to “understand” sequences of words.

Introduction to AGENT

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• An LLM agent is an AI system that uses a large language model (LLM) to perform tasks like reasoning, planning, and decision-making.



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Essay # 1 [Written by ChatGPT? __ Yes; __No]

Growing up in Casablanca, Morocco, where the hustle and bustle of the market was a male-dominated scene, I

saw how our rich traditions co-existed with undervalued roles for women. This sparked my desire to promote female empowerment. At first, I thought change should come slowly to fit within the rhythms of my culture.

My time at the Paris School of Economics was a turning point. Paris was a different world, buzzing with new ideas. Here, I met Alice, a French classmate. We often debated over cups of coffee, discussing how best to achieve gender equality. I thought slow change would work best for Morocco. But Alice said we needed fast, big changes or risk making unfair treatment of women normal.

One winter evening, Alice invited me to a panel discussion on "The Urgency of Gender Equity" held at the Sorbonne. Listening to powerful stories and evidence from around the world showed the economic and social cost of gender inequality. That night, I realized respecting culture shouldn't stop progress or justice.

Now, my fight for gender equality has a new sense of urgency. I believe in the power of education and good policy to bring real change. I hope to learn more with a Master's in Public Policy from Harvard Kennedy School. I dream of the day when Moroccan women walk freely in the marketplaces, not just as customers but as successful shop owners. I'm committed to making sure no girl or woman should have to limit her dreams because she's a woman, no matter
how hard it may be.

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Essay # 2 [Written by ChatGPT? __ Yes; __No]

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As a middle-class, suburban, American child of the digital age, I grew up with access to the latest and greatest technologies. For much of my life, I viewed technological advancement as the panacea for all socio-economic issues. My faith in technology was unshakable until a life-changing experience during a community outreach program in rural Yunnan, China while I was studying abroad.

I volunteered to teach basic digital skills to rural farmers, optimistically believing I could bridge the digital divide. The reality was startlingly different. The villagers, though polite and appreciative, seemed largely indifferent to my teachings. The smartphones and computers were alien to their traditional way of life and seemed to offer little tangible value to them.

Frustrated, I gradually fostered deeper conversations with them. I listened to their stories, shared meals, and even participated in farming activities. I saw firsthand the deep connection they had with their lands, their culture, and their community. Their way of life was vastly different, but it was rich and meaningful. This realization hit me hard, shattering my preconceived notions.

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I learned that progress and value are subjective. What I had previously viewed as backwardness was, in fact, a different, but equally valid, way of life. The experience made me reconsider the singular notion of progress I held, which was centered around technology and urban development.

Today, I seek admission to the Master of Public Policy program at Harvard Kennedy School not just to learn about policy frameworks and global issues, but to also appreciate the diversity and depth of human experiences. I hope to develop policies that respect and protect these rich cultural and community diversities. This personal transformation inspires my journey toward empathetic and inclusive policymaking.

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Essay # 3 [Written by ChatGPT? __ Yes; __No]

The event that fundamentally altered my persona was the devastating takeover of my country by the Taliban. It was as if I were a spectator at a grim theater, observing an unsettling revolution unfold in real time. I bore witness to the swift collapse of the governmental system I had grown accustomed to, and its replacement with a malevolent regime, all in the blink of an eye.

Having been raised during the democratic phase of Afghanistan, I could never have envisioned the return to power of a group mired in antiquated superstitions. This faction, responsible for over two decades of systematic brutality and terror against Afghans, destruction of infrastructure, execution of journalists and social activists, and ruthless massacre of women and children, would now impose their savage code of laws upon us. The nightmare that I had dreaded materialized before my eyes, casting me into a prolonged depression that lasted two months.

Yet, paradoxically, this disheartening episode proved to be enlightening and led to significant personal growth. The shock wave that swept through me instilled a robust sense of duty towards my fellow countrymen, particularly towards vulnerable women and children. It also fostered a resilience in me, a determination that I would endure whatever adversities life might hurl my way and continue to forge ahead.

The psychologist Victor Frankl's insights in his book, "Man's Search For Meaning," are apt in this context. He posited that while external forces might strip away all we have, they cannot take away our freedom to choose

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our responses to various situations. Having experienced what felt like the nadir of my life, akin to the deep darkness described by Fyodor Dostoevsky in his novel “Notes From Underground,” I now understand that progress is not about speed, but rather about perseverance. No matter how arduous my journey becomes, I will not cease to move forward.

Thank you !!!

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